Auction Mechanisms and Optimal Bidding Strategies

A Monte Carlo Simulation Study of Bidder Surplus and Seller Revenue

Research Portfolio Report

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Source of numerical results: Auction_Mechanisms_Research_Portfolio.xlsx (fixed-valuation simulation workbook).

Abstract

A landowner must select an **auction rule** while individual buyers decide **how aggressively to bid under uncertainty about competing valuations**. This study compares **first-price** and **second-price sealed-bid auctions** in a stylized independent-private-values environment.

Bidder valuations are independently uniform on [0, 1] million dollars; the number of participants is **N = 2, 5, or 10**. For each configuration, bidder 1 evaluates ten bid coefficients from 0.1 to 1.0 using **1,000** fixed **Monte Carlo replications**. Other bidders follow the symmetric equilibrium bid rule in the **first-price** auction and truthful bidding in the **second-price** auction.

The spreadsheet calculates **bidder-1 surplus** and **seller revenue** separately for every replication and strategy. The estimated surplus-maximizing first-price coefficients are **0.5, 0.8, and 0.9** for **N = 2, 5, and 10**; truthful bidding (**x = 1.0**) maximizes simulated second-price surplus in all three cases. **Equilibrium seller revenue increases with participation and is similar across auction formats:** for example, at N = 10 the estimates are 0.818503 and 0.817852 million dollars, respectively.

The analysis illustrates **strategic bid shading**, **competition effects**, and **revenue equivalence**, while distinguishing model-implied mechanisms from claims about real-world auction performance.

Keywords: sealed-bid auction; first-price auction; second-price auction; independent private values; Monte Carlo simulation; revenue equivalence; bidding strategy.

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1. Introduction and decision context

Consider CA, an **owner** planning to sell a single parcel of land to a group of interested buyers. CA must choose the **auction mechanism**, while each **buyer** must decide **how much to bid** without observing competitors' private valuations. Two natural formats are available. In a **first-price** sealed-bid auction, the **highest** bidder receives the land and pays their **own** bid. In a **second-price** sealed-bid auction, the **highest** bidder receives the land but pays the **second-highest** bid. The difference in payment rules changes buyers' incentives, even when the same buyer wins the parcel.

The **managerial question** is not simply which auction produces the highest observed price in a single realization. **CA** cares about **expected proceeds across uncertain buyer valuations**, whereas a participating **bidder** cares about **expected surplus after payment**. A mechanism can encourage a high submitted bid without delivering the same expected revenue advantage in equilibrium. Simulation provides a transparent way to examine this distinction at the level of individual auctions and aggregate outcomes.

This report addresses **four questions**: **(i) how does bidder 1's surplus vary with the fraction of value submitted as a bid**; **(ii) how does that strategy affect CA's expected revenue**; **(iii) how does increasing the number of bidders change the optimal strategy and outcomes**; and **(iv) which auction format is preferable under the stated assumptions?** The analysis is a computational model, not a dataset of actual land sales.

2. Theoretical framework

2.1. Information structure and bidder values

There is one indivisible parcel and **N** risk-neutral bidders. Each bidder i privately observes a valuation V_i, independently drawn from Uniform(0, 1), measured in millions of dollars. The common distribution is known, but realized competitors' valuations are not. Bidder 1 chooses a multiplicative strategy $b_1(x) = x\,V_i$ with $x \in \{0.1, 0.2, \ldots, 1.0\}$. The simulation holds other bidders' strategies fixed at the mechanism-specific benchmark to isolate bidder 1's incentives.

2.2. First-price mechanism

In the **first-price** format, the **highest** bid wins and the winner pays that bid. Opponents i = 2, …, N submit $b_i = [(N - 1)/N]\,V_i$. If bidder 1 wins, surplus is $V_1 - x\,V_1$; otherwise it is zero. Seller revenue is the maximum of all N bids. Accordingly, for any strategy x, $S_{FP}(x) = E[(V_1 - x\,V_1)\,I\{x\,V_1 \geq \max_{i\geq 2}\{b_i\}\}]$ and $R_{FP}(x) = E[\max\{x\,V_1, b_2, \ldots, b_N\}]$. With independent uniform private values and symmetric risk-neutral bidders, the equilibrium bid coefficient is $x^* = (N - 1)/N$. This coefficient rises as competition intensifies: 0.5 for two bidders, 0.8 for five, and 0.9 for ten.

2.3. Second-price mechanism

In the **second-price** format, opponents bid their values $b_i = V_i$. The **highest** bidder wins but pays the **second-highest bid**, so bidder 1's surplus is the difference between their valuation and the highest competing bid when they win. Seller revenue equals the second-highest bid across all participants. Thus

S_{SP}(x)= E[(V_1 − max_{i≥2} V_i) I{x V_1 ≥ max_{i≥2} V_i}] and R_{SP}(x) = E[second-highest{x V_1, V_2, …, V_N}]. Truthful bidding (x = 1) is weakly dominant: lowering one's bid can forgo profitable wins but cannot reduce the price paid conditional on winning. The logic follows the second-price incentive structure analyzed by Vickrey (1961).

2.4. Revenue equivalence as a benchmark

Under symmetric equilibrium and the model's regularity assumptions, the two formats yield the **same expected seller revenue**. With N independent Uniform(0, 1) valuations, the expected second-highest valuation is **(N − 1)/(N + 1)** million dollars. Hence the theoretical benchmark is 1/3 for N = 2, 2/3 for N = 5, and 9/11 for N = 10. Equality concerns expectations in equilibrium, not identical revenues for each realized auction or for arbitrary off-equilibrium strategies.

3. Simulation design and implementation

3.1. Experimental factors and estimands

Design element	Specification
Auction formats	First-price and second-price sealed bid
Bidder counts	N = 2, 5, 10
Bidder-1 strategies	x = 0.1, 0.2, …, 1.0
Monte Carlo sample	K = 1,000 replications per bidder-count configuration
Private valuations	Independent Uniform(0, 1) in $M
Primary outcomes	Expected bidder-1 surplus; expected seller revenue
Comparability	Same fixed valuation draws across strategies and mechanisms for each N

The experiment estimates E[S_1(x)] and E[R_N(x)] by averaging 1,000 simulated outcomes for each of ten strategies and each auction format. For a given N and replication, the same valuation vector is reused across alternative values of x. The two auction formats also share those valuations. This common-random-numbers design helps distinguish the effect of the auction rule and bid coefficient from differences in simulated market conditions. The three bidder-count scenarios have separate fixed valuation tables.

3.2. Sample path, strategy grids, and summary statistics

Each **sample path** records an individual auction realization: bidders' valuations, their bids under the specified rule, the winner, the applicable payment, bidder 1's realized surplus, and seller revenue. A **strategy grid** then evaluates ten bidding coefficients for 1,000 replications. One grid stores **bidder surplus** and another **seller revenue** for each N × mechanism combination, producing twelve 1,000-by-10 simulation matrices. Column means summarize **expected surplus** and **expected revenue**. The workbook also retains sample-path calculations and explanatory documentation so readers can trace the outcome formulas back to the auction rules.

First-price seller revenue is the maximum submitted bid. **Second-price seller revenue** is the second-highest submitted bid; the spreadsheet implements this with an order-statistic equivalent that accounts for bidder 1 and the competing bidders. These distinct definitions are essential: treating second-price revenue as the maximum bid would overstate proceeds and invalidate the comparison.

3.3. Workbook structure and reproducibility

The accompanying workbook separates research narrative, model assumptions, design, results, strategy curves, convergence, interpretation, implications, and calculation guidance from the two mechanism-specific calculation sheets. The valuation tables MC_Random_N2, MC_Random_N5, and MC_Random_N10 contain fixed draws, so recalculation does not create a new Monte Carlo sample. The reported numbers are taken from the workbook's saved strategy summaries and were cross-checked against the corresponding 1,000-replication matrices. The report's plots are rendered directly from those same workbook data; no alternative valuations were generated.

4. Simulation results

4.1. Optimal bid coefficients and bidder surplus

Bidders N	FP optimal x	FP surplus ($M)	SP optimal x	SP surplus ($M)
2	0.5	0.164415	1.0	0.164690
5	0.8	0.033928	1.0	0.033918
10	0.9	0.010406	1.0	0.010577

The **first-price surplus** estimates peak exactly at the equilibrium shading coefficients: **0.5, 0.8,** and **0.9**. Bidder 1 trades off a greater chance of winning against a smaller surplus conditional on winning. At **x = 1**, a first-price winner pays their entire valuation and therefore receives zero surplus, even when the bid wins. Under **second-price** rules, the simulated **surplus increases toward x = 1 in all three scenarios; truthful bidding avoids rejecting opportunities where the valuation exceeds the price that would be paid**.

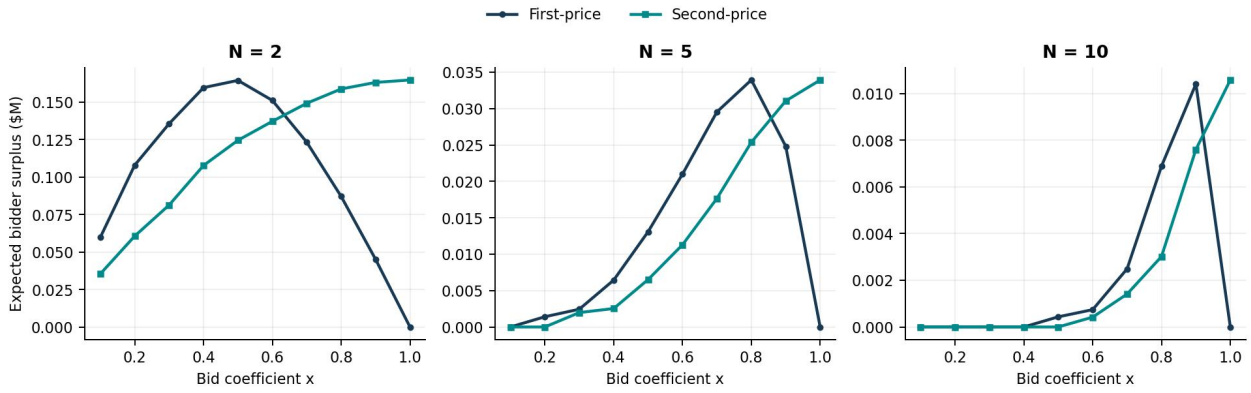


Figure 1. Bidder-1 expected surplus by bidding coefficient and auction mechanism.

Each point is the mean of 1,000 fixed simulation outcomes.

The levels of optimal surplus fall as the number of competitors grows. At **N = 2**, estimated optimal first-price surplus is 0.164415 million dollars; at **N = 10** it is 0.010406 million dollars. This is economically intuitive: **with more bidders, the highest rival valuation tends to be larger, so any one bidder wins less often and faces more competition for the remaining gains from trade**.

4.2. Seller revenue across strategies

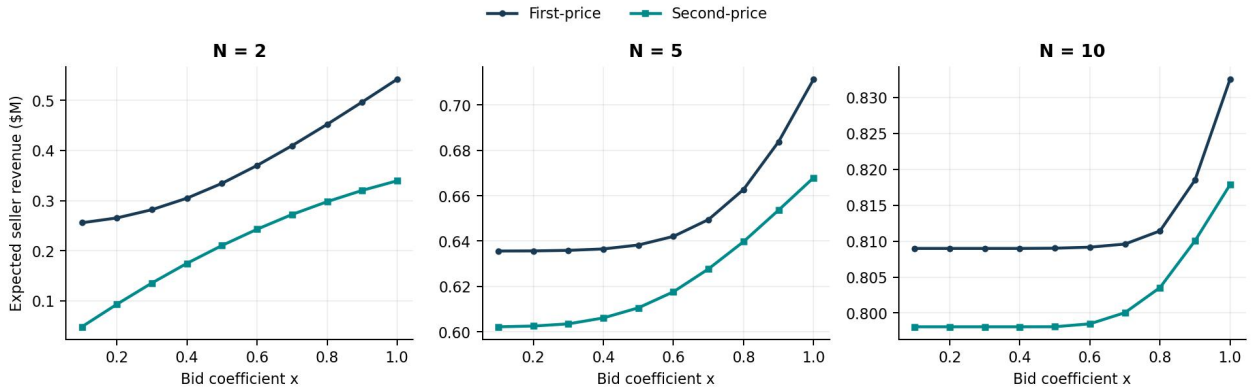


Figure 2. Expected seller revenue by bidder-1 bid coefficient for N = 2, 5, and 10.

Values are means from the workbook's revenue grids.

Revenue generally increases as bidder 1 submits a higher fraction of their valuation. This pattern must not be confused with bidder 1's optimal behavior. For example, in the first-price N = 2 scenario, the workbook records seller revenue of 0.334320 million dollars at the bidder's equilibrium coefficient x = 0.5, compared with 0.541866 million dollars at x = 1.0. The latter is an off-equilibrium scenario in which bidder 1 bids more aggressively than is individually optimal; CA cannot assume the bidder will voluntarily select it. Seller revenue is a function of the auction rule and the equilibrium response of participants, not merely a strategy imposed on them.

4.3. Equilibrium revenue and theoretical comparison

N	Theory ($M)	First-price ($M)	Second-price ($M)	SP − FP ($M)
2	0.333333	0.334320	0.339390	+0.005070
5	0.666667	0.662739	0.667931	+0.005192
10	0.818182	0.818503	0.817852	-0.000650

The expected seller revenue increases sharply with the number of bidders, from approximately one-third of a million dollars with two bidders to slightly above 0.81 million dollars with ten. **The two auction formats are close at each bidder count**. The signed second-minus-first differences are +0.005070, +0.005192, and −0.000650 million dollars for N = 2, 5, and 10, respectively. Because the sign changes across scenarios and the differences are small relative to the level of revenue, these outcomes do not establish that one format systematically dominates the other.

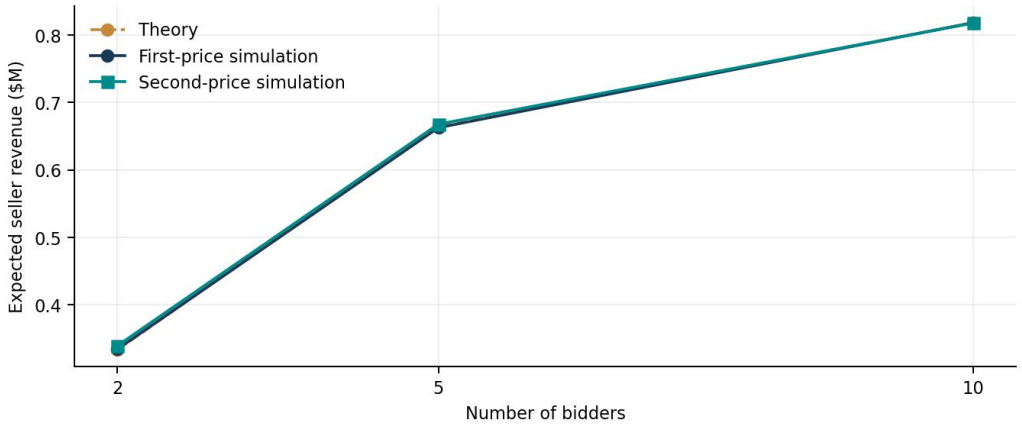


Figure 3. Equilibrium seller revenue compared with the uniform-private-values theoretical expectation (N − 1)/(N + 1).

4.4. Monte Carlo convergence and precision

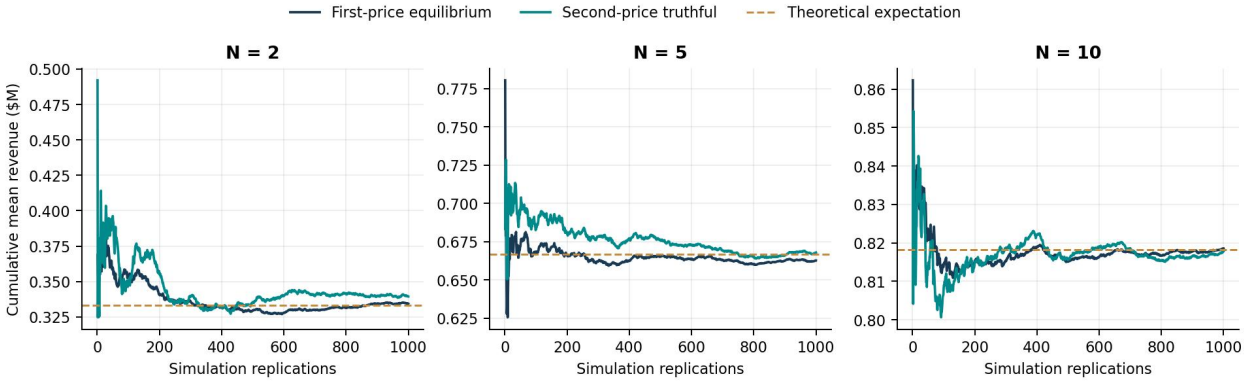


Figure 4. Running average of seller revenue at each mechanism's equilibrium strategy, calculated from the fixed 1,000-replication revenue paths. Dashed lines show theoretical expected revenue.

The running means exhibit relatively large early fluctuations when only a small number of simulated auctions have been observed. As replications accumulate, the trajectories become more **stable** and remain close to the theoretical benchmarks. This visual convergence is consistent with the law of large numbers, but it does not establish exact equality or remove Monte Carlo error. Because the valuation sample is fixed, repeated recalculation of this workbook reproduces the same trajectories rather than constituting an independent replication of the study.

N	Mechanism	Mean ($M)	MC standard error	Approx. 95% MC interval
2	First-price	0.334320	0.003714	[0.327041, 0.341599]
2	Second-price	0.339390	0.007396	[0.324895, 0.353886]
5	First-price	0.662739	0.003691	[0.655505, 0.669973]
5	Second-price	0.667931	0.005539	[0.657075, 0.678787]
10	First-price	0.818503	0.002266	[0.814062, 0.822944]
10	Second-price	0.817852	0.003407	[0.811174, 0.824531]

The standard errors above are calculated from the fixed revenue paths using the sample standard deviation divided by √1,000; the displayed intervals use the normal approximation mean ± 1.96 × standard error. These are Monte Carlo uncertainty summaries conditional on the model and the selected strategies, not confidence intervals for observed real-estate market outcomes. The revenue differences between mechanisms should ideally be assessed with a paired standard error because the two simulations share valuation draws; unpaired interval overlap alone is not a test of mechanism equivalence.

5. Interpretation and discussion

5.1. Why optimal bidding differs across mechanisms

First-price bidders internalize the cost of every additional dollar they bid if they win. **Shading** therefore protects surplus but reduces the probability of allocation. The equilibrium fraction $(N - 1)/N$ balances those incentives under the specified distribution and rival behavior. **Second-price** bidders do not determine their payment by their own winning bid: their bid determines whether they win, while the second-highest rival bid determines what they pay. Consequently, **truthful bidding** is weakly dominant in the modeled second-price auction. The contrast is a consequence of payment design, not an inconsistency in bidder rationality.

5.2. Competition and the distribution of surplus

As N rises, the highest competing valuation tends to rise. The simulated **optimal bidder surplus shrinks substantially while seller revenue rises**. This is a distributional shift in the benefits from the auction: **competition transfers a greater share of the available gains from trade toward the seller**. The analysis does not measure total welfare directly in the summary tables, but under symmetric equilibrium both mechanisms allocate the object to the bidder with the highest valuation almost surely, consistent with allocative efficiency in this model.

5.3. What revenue equivalence does—and does not—say

Revenue equivalence is a statement about expected proceeds under specified equilibrium and information assumptions. It does not imply identical bid distributions, identical winner payments in individual auctions, or identical revenue when bidder 1 deviates from equilibrium. The small observed discrepancies arise from the finite Monte Carlo sample and the different realized payment mappings. The close agreement with the order-statistic benchmark supports the internal consistency of the implemented mechanisms, but does not prove that revenue equivalence holds under every institutional design.

6. Managerial and analytical implications

For **CA**, the primary lesson is that **choosing between standard first-price and second-price sealed-bid rules solely to raise expected revenue is not justified by this benchmark model.** Expected proceeds are nearly equivalent when bidders behave according to the assumed equilibrium strategies. Instead, the practical choice may depend on implementation and institutional considerations not included in the simulation: bidder understanding, disclosure and transparency, enforceability, susceptibility to collusion, and operational simplicity. Such considerations require separate evidence rather than extrapolation from the modeled revenue figures.

For prospective **bidders**, **the optimal strategy depends on the payment rule**. In a **first-price** setting, bidding one's full valuation sacrifices all surplus on wins; **equilibrium shading** responds to the intensity of competition. In a **second-price** setting with independent private values, **truthful bidding** protects the bidder from foregoing beneficial allocations without increasing the conditional payment. These prescriptions should not be transferred uncritically to common-value auctions or settings with budgets, correlated signals, or risk aversion.

Analytically, the **project** demonstrates the value of simulation as a bridge between a theoretical game and **decision-relevant outputs**. Using identical valuation draws across policy alternatives reduces avoidable comparison noise; separating bidder utility from seller revenue prevents objective-function confusion; and convergence plots and standard errors communicate the precision of estimates. The workbook's traceable sample paths and strategy grids make these results auditable at the level of individual realizations.

7. Limitations and future extensions

The study assumes independent Uniform(0, 1) valuations, risk neutrality, symmetric information about the distribution, a single indivisible asset, and exogenously fixed opponent strategies. These assumptions support tractable theoretical benchmarks but exclude important real-market phenomena such as valuation correlation, common-value uncertainty, entry costs, financing constraints, collusion, reserve prices, asymmetric bidders, and endogenous participation. Results should therefore be read as properties of the specified model, not forecasts for an actual land auction.

The bid grid is discrete, with increments of 0.1. This is sufficient to include the predicted equilibrium coefficients for N = 2, 5, and 10, but a finer grid would be needed to study arbitrary parameter settings or more subtle local deviations. The 1,000 replications provide a useful demonstration but still leave sampling uncertainty, especially for the relatively rare event of bidder 1 winning when N is large. The report includes convergence diagnostics and Monte Carlo standard errors, but does not claim a formal power analysis or exhaustive robustness assessment.

Future work could vary the valuation distribution, increase replications, introduce reserve prices or bidder heterogeneity, evaluate paired uncertainty for differences in seller revenue, and examine the trade-off between revenue, allocative efficiency, and bidder participation. These are extensions, not analyses already performed in the accompanying workbook.

8. Conclusion

The simulation **demonstrates how auction payment rules shape bidder incentives even when both mechanisms select the highest bidder**. **First-price equilibrium bidding involves shading one's valuation**, with the optimal fraction **increasing** from 0.5 to 0.9 as competition expands from two to ten bidders. **Second-price truthful bidding remains optimal across all three bidder-count scenarios. Seller revenue increases** with the number of bidders, while the **bidder's expected surplus declines**. At equilibrium, the two mechanisms produce closely aligned expected revenue and track the theoretical order-statistic benchmark.

For a seller deciding between these formats under the specified assumptions, mechanism choice should not rest on a presumed systematic revenue advantage; for analysts, the larger contribution is a **transparent, reproducible comparison** linking **individual bidding behavior, auction outcomes,** and **Monte Carlo uncertainty.**

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Appendix A. Workbook Structure

Workbook sheet	**Role in this report**
1_First_Price	Sample-path and strategy-by-replication surplus/revenue calculations
2_Second_Price	Second-price calculations and mechanism comparison
MC_Random_N2 / N5 / N10	Fixed common valuation draws
03_RESULTS_DASHBOARD	Summary of optimum strategy and equilibrium comparisons
04_STRATEGY_CURVES	Ten-strategy profiles and mechanism comparison
05_CONVERGENCE	Replication-based stability and uncertainty diagnostics
08_CALCULATION_GUIDE	Definitions of sample paths, data tables, statistics, and questions

Fixed Numbers: all twelve 1,000 × 10 outcome matrices were extracted from the saved calculation sheets and their column means were checked against the corresponding workbook summary cells before figure creation. Report figures use the same saved values. Monetary values are expressed in millions of dollars; rounding in tables does not affect underlying calculations.